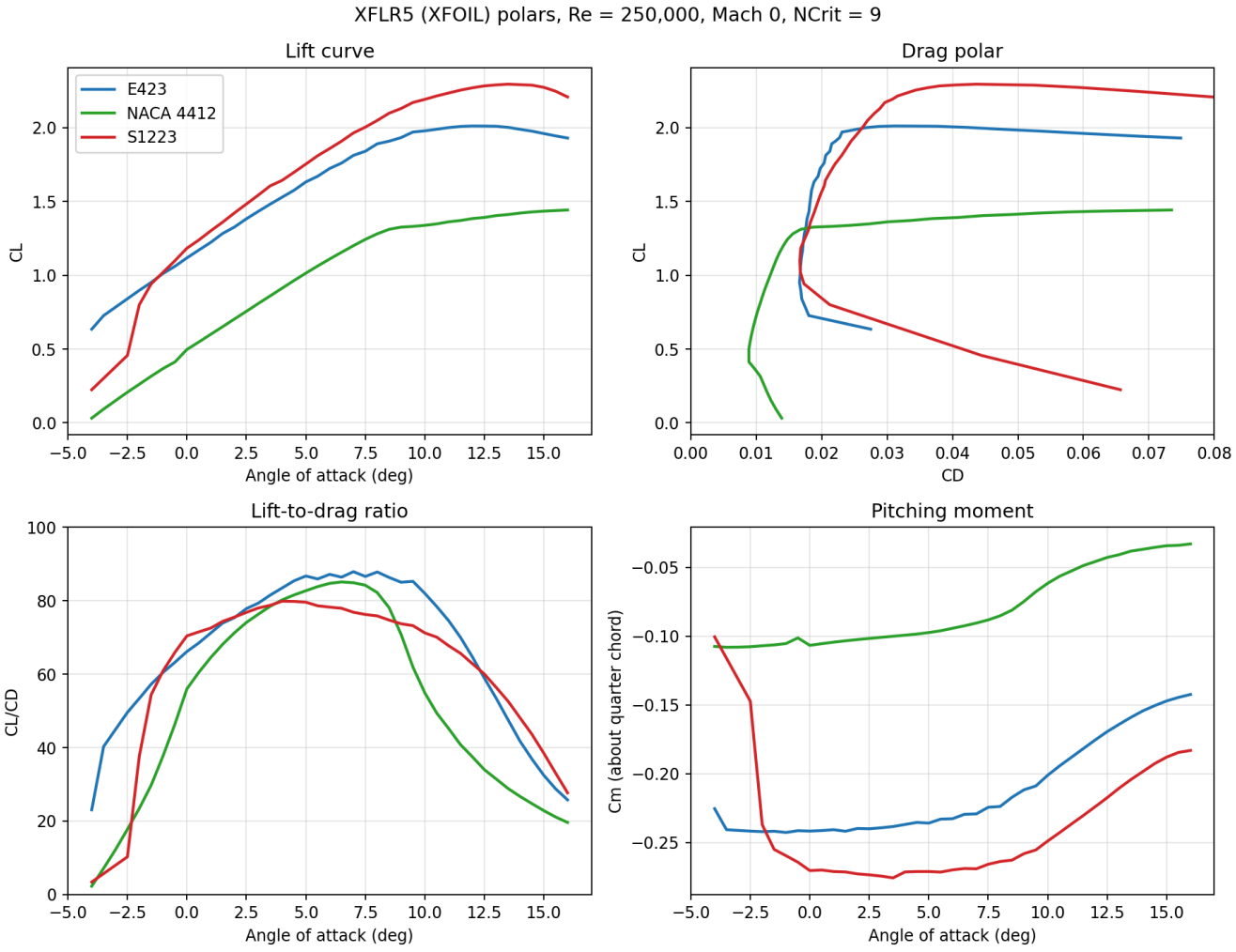
Airfoil Comparison for Aircraft A: NACA 4412, E423 and S1223

Results report from XFLR5 (XFOIL) polars. Figure: see the attached image airfoil_polars_Re250k.png (four plots: lift curve, drag polar, lift-to-drag ratio, pitching moment).



Results Table

All values are read from your exported files. Camber and thickness are approximate catalogue values and were not part of the files.

Quantity	NACA 4412	E423	S1223
Thickness / camber (approx.)	12% / 4%	12.5% / 9–10%	~12% / ~8.7%
Maximum lift, CLmax	**≥1.44** (still rising at 16°)	**2.01** at 12.0°	**2.29** at 13.5°
Lift at 0° angle, CL	0.50	1.12	1.18
Zero-lift angle (extrapolated)	−4.8°	−10.7°	−10.0°
Lowest drag, CD min	0.0089	0.0166	0.0167
Drag at CL = 1.0	0.0121	0.0167	0.0169
Best lift-to-drag, (L/D)max	**85** at CL 1.16	**88** at CL 1.81	**80** at CL 1.64
L/D at CL = 1.0	82	60	59
Pitching moment, Cm (nose-down is negative)	**−0.107**	**−0.242**	**−0.270**
Lift lost between peak and 16°	none (no peak yet)	4.1%	3.8%
Stall behaviour	Very gradual; lift curve bends at about 9° (CL 1.33)	Gradual, rounded peak	Gradual, rounded peak

How to read it: CL is lift, CD is drag, and L/D is how much lift you get per unit of drag. A higher CLmax means the wing can carry more weight at a slow speed. A more negative Cm means a stronger nose-down twist that the tail must balance.

What It Means for Aircraft A

Sample aircraft (assumed): wing area 0.55 m², weight 58.9 N (6 kg), cruise 14 m/s, wing chord 0.275 m, tail arm 0.8 m. The real wing reaches about 87% of the 2D CLmax (assumed rule of thumb).

Quantity	NACA 4412	E423	S1223
Usable wing CLmax	1.25	1.75	2.00

Quantity	NACA 4412	E423	S1223
Stall speed	11.8 m/s	10.0 m/s	9.4 m/s
Nose-down moment at 14 m/s	1.9 N·m	4.4 N·m	4.9 N·m
Tail download needed	2.4 N (4% of weight)	5.5 N (9%)	6.1 N (10%)

The tail download is a downward push that the tail must make to cancel the nose-down twist. It uses up lift, so it eats into payload. This is a simplified estimate.

Discussion

- **E423 vs. NACA 4412:** E423 carries about 40% more lift at stall (2.01 vs. 1.44), so the aircraft can stall about 1.8 m/s slower or carry more weight. Its best L/D (88) is also slightly higher, and it occurs at a very high CL (1.81), which is useful for a heavy aircraft. The price is more drag at low lift (0.0166 vs. 0.0089) and a larger nose-down moment (about 2.3 times).
- **S1223 vs. E423:** S1223 gives 14% more peak lift (2.29 vs. 2.01) and a stall speed about 0.6 m/s lower. But its best L/D is lower (80 vs. 88), it needs about 11% more tail download, and the gain in stall speed is small. E423 is the better balance of lift, efficiency and stability.
- **Stall:** E423 and S1223 both have rounded lift peaks, which suggests a gradual stall. NACA 4412 has an even softer one, since lift keeps creeping upward past 9° (with a sharp rise in drag there). These are 2D results, so confirm in flight tests.

Data Quality and Limits

- **Missing and unreliable points:** E423 has no results at −3.0° and 4.0°. S1223 is missing several points, including 14.0°. Both show high drag and odd values at the lowest angles (−4° and −2.5°), probably from air separation bubbles or an unconverged run, so those points were ignored.
- **Drag at CL = 0.5 is not reported** for E423 and S1223, because the converged results do not cover that lift level. Drag at CL = 1.0 is used instead.
- **NACA 4412 did not reach its peak.** The sweep stopped at 16° while lift was still rising, so ≥1.44 is a lower bound. Re-run to about 20°. For safety use the knee value (CL about 1.33 at 9°).

- **XFOIL is optimistic and post-stall values are unreliable.** A real model has rougher surfaces and more turbulence, so real drag will be higher and real lift lower. The comparison between airfoils is more trustworthy than any single number.

Comparison With the Earlier Estimates

The approximate table in the earlier report was based on typical values. The real polars changed a few things:

- E423 CLmax (2.01) and Cm (−0.24) agree with the estimates. Its best L/D (88) is higher than the estimated 65–80.
- S1223 CLmax (2.29) is slightly above the estimated 2.1–2.2. Cm (−0.27) agrees.
- NACA 4412 best L/D (85) is far above the estimated 45–55, and CLmax is at least 1.44. XFOIL at this Reynolds number is optimistic for thin, smooth sections.

Update the earlier comparison table with these numbers.

Conclusion

E423 is still the best choice for the payload aircraft: it provides high lift (CLmax 2.01), the best peak L/D (88), a gradual stall, and a manageable nose-down moment. S1223 offers more lift but costs efficiency and needs a larger tail load. NACA 4412 is gentle and easy to trim but cannot carry as much weight. These results should be confirmed by flight tests, since XFLR5 results tend to be optimistic.